Experiment - 2

Name: Vinay Tiwari

UID: 24BAI7919
Section: 24AIT_KRG-1A

Aim:-

To study and implement advanced SQL queries using conditional logic, aggregate functions, joins, subqueries, self-joins, and date operations to solve real-world database problems

Question 1)

A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments. The solution must join the student profile, course subjects, and grade marks tables, returning a breakdown sorted by the highest average marks down to the lowest.

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Answer

```
Select d.dept_name as Department,
round(avg(m.marks_scored), 2) as "Average Marks"
from Departments as d
join students as s
on d.dept_id =s.dept_id
join marks as m
on m.student_id=s.student_id
group by d.dept_name
order by "Average Marks" Desc
```

Output

	department character varying (50)	Average Marks numeric
1	Mathematics	95.00
2	Computer Science	83.33

Question 2)

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to **merge these datasets** and identify **each unique employee** (by EmpID) along with their lowest recorded salary across both systems.

Objective:

• Combine two tables A and B

• Return each EmpID with their lowest salary, and the corresponding Ename

Answer

```
select C.empid, min(c.ename), min(c.salary) from
(select * from A
union all
select * from B) as C
group by c.empid
```

Output

	empid integer	min text	min integer
1	3	CC	100
2	2	BB	300
3	1	AA	1000

Question 3)

A multinational organization maintains separate records for employee information and department details. The management wants to identify the employees who receive the highest salary within their respective departments for an annual performance report.

If more than one employee earns the same highest salary in a department, all such employees should be included in the report. The final output should display the **Department Name**, **Employee Name**, and **Salary**, arranged in ascending order of department name.

Answer

```
select d.dept_name, e.name, e.salary from Employee as e
join department as d
on d.id=e.department_id
where e.salary = (select max(e2.salary) from employee as e2
where e2.department_id=d.id)
```

Output

	dept_name character varying (50)	name character varying (50)	salary integer
1	IT	JIM	90000
2	SALES	HENRY	80000
3	IT	MAX	90000